

Everything fills, and this is exactly how.

Five constructions carry every animation in the brand: the mark, the button, the type, the product and the flood. Each one is a wave-topped body rising behind a cut-out. The numbers below are the site's own — same geometry, same easing, same durations — so film and screen cannot drift apart.

SOURCE: ASSETS/CSS/BRAND.CSS MARK VECTOR: ASSETS/IMG/ZINSTIM-MARK.SVG

ARTWORK: ASSETS/IMG/ZINSTIM-MARK.PNG

Ground rules

Frame rate

Build at **60 fps**. Every duration below is given in seconds and in frames at 60 and at 30. Deliver 60 for screens, 30 only when a platform demands it — and re-time by seconds, never by frame count.

Scale

The site's shapes are drawn in SVG units. Work at **8×**: one SVG unit is 8 px. The mark's 128 × 114 box becomes a **1024 × 912** comp, and every wave number below scales with it.

The one rule

A liquid moves **only while it is being filled**. The drift and the turn run during the fill and stop when it settles. Nothing in this brand loops on its own.

Easing

Two curves do all the work. **Arriving** is used for every fill; **moving** is only for something travelling between two known positions.

CURVE	CSS	IN AFTER EFFECTS
Arriving	<code>cubic-bezier(.16, 1, .3, 1)</code>	Out influence 16% , in influence 70% with incoming speed 0 . Fast departure, long settle.
Moving	<code>cubic-bezier(.62, .03, .2, 1)</code>	Out influence 62% (slow start), in influence 80% , incoming speed 0.
Drift and turn	<code>linear</code>	No easing at all. Convert both keyframes to Linear or the loop will stutter at the seam.

Influence values get you close. For an exact match, drive the property with the expression instead — it is the same curve the browser solves.

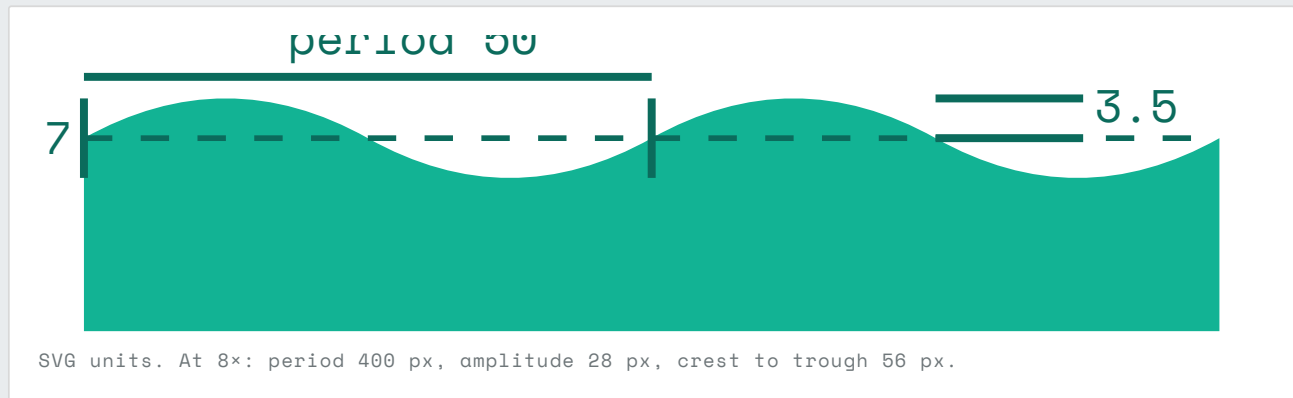
The ease, as an expression

```
// ZinStim arriving ease. Paste on any property.
// Set t0/t1 (seconds) and v0/v1 (from / to).
var t0 = 0, t1 = 0.75;
var v0 = 912, v1 = -102;           // example: the mark's level
function bez(p1x, p1y, p2x, p2y, x){
  var cx = 3*p1x, bx = 3*(p2x-p1x) - cx, ax = 1 - cx - bx;
  var cy = 3*p1y, by = 3*(p2y-p1y) - cy, ay = 1 - cy - by;
  function sx(t){ return ((ax*t + bx)*t + cx)*t; }
  function sy(t){ return ((ay*t + by)*t + cy)*t; }
  function dx(t){ return (3*ax*t + 2*bx)*t + cx; }
  var t = x;
  for (var i = 0; i < 8; i++){
    var d = sx(t) - x;
    if (Math.abs(d) < 1e-5) break;
    var s = dx(t);
    if (Math.abs(s) < 1e-6) break;
    t = t - d/s;
  }
  return sy(t);
}
var x = clamp(linear(time, t0, t1, 0, 1), 0, 1);
v0 + (v1 - v0) * bez(.16, 1, .3, 1, x)
```

DURATION	SECONDS	FRAMES @60	FRAMES @30	WHERE IT IS USED
Fast	0.18 s	11	5	Colour answering a pointer
Base	0.32 s	19	10	Small state changes, discs, pills
Label flip	0.30 s, 0.20 s late	18, starts at 12	9, starts at 6	Button text turning over as the liquid passes
Slow	0.58 s	35	17	Panels, sheets
Mark fill	0.75 s	45	23	The nav mark, and the button's blob
Type fill, hover	0.80 s	48	24	A word filling under the cursor
Vessel fill	1.05 s	63	32	A product filling inside its outline
Flood	1.10 s	66	33	A card or tile flooding
Type pour, on load	2.20 s, 0.55 s late	132, starts at 33	66, starts at 17	A page title pouring to its resting level
Wave drift	1.60 s per loop	96	48	The surface travelling sideways
Blob turn	5 s per turn	300	150	The button's body rotating
Flood turn	12 s per turn	720	360	The card's body rotating

The wave

One wave shape sits on top of every liquid in the brand. Build it once, reuse it everywhere.



Geometry

Period	50 units · 400 px at 8×
Amplitude	±3.5 units · ±28 px
Crest to trough	7 units · 56 px
Baseline	y = 7 units from the top of the wave strip
Curve	Quadratic, control point on the baseline — a sine, not a scallop
Body	Solid fill from the wave down; the body is never a gradient on a surface

The path, to paste into Illustrator

```
M0 7 Q12.5 0 25 7 T50 7 T75 7 T100 7 V14 H0 Z
```

Paste as a path, then **Layer** → **Create Shapes from Vector Layer** in After Effects.

Build the strip at least **3× the width of the vessel** so it never runs out while drifting.

Drift: the seam rule

The surface travels sideways **exactly two periods — 100 units, 800 px at 8×** — every **1.6 s, linear, and loops**. Any other distance leaves a visible jump at the loop point. Use the expression rather than keyframes and it can never drift out of phase:

```
// X position of the wave strip. 800 px = 2 periods at 8×.
var per = 1.6, dist = 800;
value - (time % per) / per * dist
```

The drift runs **while the level is moving** and for about two passes after it settles, then stops. On the site the resting state is still.

The mark reveal

A liquid rises inside the mark's silhouette and uncovers the carbon-fibre artwork.
Two cut-outs, one moving body: the level is the only thing that animates.

Comp and cut-outs

Comp	1024 × 912 px · 60 fps
Source box	128 × 114 SVG units at 8×
Cut-out 1	The mark silhouette — blade and wedge, one closed shape, from <code>zinstim-mark.svg</code>
Cut-out 2	The wave body, which is what moves
Artwork	<code>zinstim-mark.png</code> , placed to fill the 1024 × 912 box exactly — never scaled to taste
Rest colour	Zinc Silver #D6DDE2 on Night, Obsidian #0D0D0D on Day

Layer stack, top to bottom

1 MARK-SILHOUETTE · white fill · set as Alpha Matte

2 REVEAL (precomp) — the artwork, matted by the liquid

inside REVEAL:

a LIQUID · wave strip · Alpha Matte

b ARTWORK · `zinstim-mark.png`

back in the main comp:

3 MARK-REST · the same silhouette, flat Zinc Silver

The silhouette is used twice for the same reason the site uses it twice: once to clip, once to draw. The liquid can then never leave the mark, and the edge stays crisp over it.

The movement

FRAME @60	LIQUID Y	LEVEL	NOTE
0	+912 px	0	Wave crest sits on the bottom edge, nothing showing
45	-109 px	1.12	Full. Arriving ease on both keyframes
hold	-109 px	1.12	As long as the cursor stays. For film, 0.5–1 s
+45	+912 px	0	Drains at the same speed when the cursor leaves

Why 1.12 and not 1. At level 1 the wave's troughs are still 10.5 units — 84 px at 8×

below the top of the mark, so a row of base-coloured bumps rides along the blade.

Measured on the site: 356 changed pixels at 8 rem before the fix, 0 after. Overshoot to 1.12 and the crest clears the edge.

Set-up notes

- ☐ Liquid strip: 4800 px wide, body 1920 px deep, so it covers the mark at every level and through the whole drift.

- ☐ Drift the strip -800 px per 1.6 s, linear, only while the level is moving.
- ☐ Anchor the strip's *wave line*, not its box, at $Y = 0$ so the numbers above read directly.
- ☐ Motion blur off. The liquid is a surface, not a moving object.
- ☐ The artwork never moves. Only the edge travels — that is what makes it read as a reveal instead of a wipe.

The liquid button

A large turning square with soft corners rises through the pill. Its curved edge is the meniscus; the label turns over as the surface passes it.

Geometry, as a formula

PART	RULE	WORKED EXAMPLE
Button	Any size	176 × 46 px
Blob	Square, side = max(2.4 × button width, 224 px)	422 × 422 px
Corner radius	$0.42 \times \text{side}$	177 px
Start, top edge	+118% of button height	+54 px
End, top edge	-46% of button height	-21 px
Travel	164% of button height	75 px
Horizontal	Centred on the button, and it stays centred	x = 88 px

Work at 4× for film: the example button becomes a 704 × 184 px comp and every number multiplies by four.

The movement

FRAME @60	WHAT	VALUE
0	Blob top edge	+54 px · arriving ease
45	Blob top edge	-21 px
0 → 300	Blob rotation	0° → 360°, linear, looping
12 → 30	Label colour	Zinc Wash → Obsidian, 0.3 s, plain ease
on press	Whole button	scale 100% → 98.5%, 0.18 s
on exit	Blob	Back to +54 px, same 45 frames, rotation stops

- ☐ Clip everything to the pill: put the pill shape above as an Alpha Matte, or mask the precomp.
- ☐ Rotate about the blob's own centre, not the button's.
- ☐ The blob is flat colour. No gradient, no blur, no glow inside the pill.
- ☐ film only Matte the second label copy with the blob so the text flips pixel by pixel as the surface crosses it. Handsome on film; the site flips the whole word on a timer, and that stays.

Which colours

On Night: the pill is crimson #AE4047 with a Zinc Wash label, and it fills with #F29BA1 as the label turns Obsidian. On Day: the pill is Obsidian with a Zinc Wash label, and it fills with Elixir #0B6B5C — the label does not change, because Zinc Wash reads 5.4:1 on the elixir. Secondary buttons are an outline that fills with 16% Zinc Wash.

Liquid type

The word is drawn twice: solid type underneath, and a wave-topped body on top clipped to the letterforms. The empty part is never a faint ghost — it is full-strength type.

Build

- 1 WORD · the text layer · Alpha Matte
- 2 FILL (precomp) – flat colour with the wave on its top edge
- 3 WORD · the same text, solid, full strength

Layer 1 and layer 3 are duplicates: same face, size, tracking and position. Archivo at width 78, weight 900 for display. If they drift by a pixel the fill will look like a shadow.

The box

Liquid box height	1.32 em of the type size
At 152 px type	200 px
Wave height	0.15 em · 23 px at 152 px
Wave overlap	0.14 em above the waterline
Drift distance	0.9 em per pass

Levels and timing

MOMENT	LEVEL	TIMING
Page title, on load	0 → 0.5	2.2 s, starting 0.55 s in · 132 f
Under the cursor	0.5 → 1	0.8 s · 48 f
Cursor leaves	1 → 0.5	0.9 s · 54 f
On a phone	0.5, still	No hover: the idea has to be visible at rest
Drift, settling	—	1.4 s per pass, two passes, then still
Drift, while held	—	1.1 s per pass, looping

Why 0.5 is the resting level. Measured, not guessed: at 152 px the cap height is 0.691 em inside the 1.32 em box, so 0.5 puts the waterline 51% down the ink — a real half.

The first guess, 0.6, sat 32% down and the words read as colour with dark smudges on top.

Waterline Y inside the box = box × (1 - level) - 0.14 em.

The product vessel

A garment or a tub fills inside its own outline. Same construction as the mark, different cut-out.

Build and numbers

Cut-out	The product outline, closed — used as the matte
Line on top	2.4 units · 19 px at 8×, round joins
Second line	same path at 55% opacity, for the seam
Fill	1.05 s · 63 f · arriving ease
Drift	-800 px per 1.6 s while filling

Levels

PIECE	REST	POINTED AT	ON A PHONE
A tee in the drop	0.10	0.94	0.50
The Fuel tub	0.16	0.90	0.50
The mark	0	1.12	Opens with the menu

A product never fills to 1: a full garment reads as a solid shape and the drawing disappears. 0.94 leaves the shoulders as line.

The flood

A whole card or category tile fills. The same turning body as the button, scaled to the card and slowed down.

PART	VALUE
Blob	280% of the card width, square
Corner radius	$0.43 \times \text{side}$
Start, top edge	+104% of card height
End, top edge	-18% of card height
Fill	1.1 s · 66 f · arriving ease
Turn	360° per 12 s, linear, while filling
Small type on the card	brightens over 0.4 s, 0.3 s late

Notes

- ☐ Clip to the card, including its corner radius.
- ☐ The flood colour is flat — Deep crimson #480F15 or Elixir deep #0A4A3F. A gradient on a turning shape turns with it, which is the tell.
- ☐ Small grey type fails on the flood, so the specs brighten to Zinc Silver as the surface reaches them. That colour change is information, not decoration — keep it.

Export, and what not to do

Deliver

USE	FORMAT
Master	ProRes 4444 with alpha, 60 fps, at 2× the largest place it will sit
Review	H.264, 60 fps, same frame size
Web overlay	PNG sequence with alpha, or WebM with alpha
Interface	Do not export. The site builds these in CSS — film is for campaign, not for the button.
Stills	Rest frame and full frame of every construction, for the brand book

Name files `zinstim_[construction]_[state]_[fps]_[size].[ext]` — for example `zinstim_mark_reveal_60_2048.mov`.

If it has to be Lottie

- ☐ Text clipped to a fill has no Lottie equivalent: rebuild the word as outlines and matte it.
- ☐ Keep everything as shape layers. No effects, no track mattes on precomps that Bodymovin will flatten.
- ☐ Re-check the loop seam after export: the drift must still travel exactly two periods.
- ☐ never Auto-trace the mark. Use the supplied vector — the traced version misses the wedge's inner angle.

BEFORE YOU SEND IT

The mark's level overshoots to 1.12 — no bumps riding along the top edge at full.

The drift travels exactly two periods per loop. Scrub the seam frame by frame.

Drift and turn are linear; every fill uses the arriving ease.

Nothing moves once the fill has settled.

The liquid never crosses the cut-out — check the blade tip and the garment hem at full.

The button label turns over 0.2 s after the fill starts, over 0.3 s.

Flat colour everywhere. Gradients only inside a pour, never on a surface.

A still of the finished state exists, for reduced motion and for print.